STAFF REVIEW · H446 017 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. 3 and 3. Dimension 0 is rows; dimension 1 is columns.

2. A function returns a value; a procedure does not.

3. 47. 2 × 16 + 15 = 47 .

4. An array has one declared element type, so every element must be compatible with that type.

I Do — source worked steps

1. Rows are students · columns are fields — CODE:
row 0 → Amelia | 72
row 1 → Ben | 65
row 2 → Charlie| 81 — We need 3 rows and 2 columns.

2. Create the 2D array — CODE:
string[,] studentResults =
{
{ "Amelia", "72" },
{ "Ben", "65" },
{ "Charlie", "81" }
}; — Both columns use strings in this teaching example.

3. Use row, then column — CODE:
Console.WriteLine(studentResults[0,0]);
Console.WriteLine(studentResults[0,1]); — The outputs are Amelia and 72 .

4. Visit every row — CODE:
for (int row = 0;
row < studentResults.GetLength(0);
row++)
{
Console.WriteLine(
studentResults[row,0] + " " +
studentResults[row,1]);
}

5. Think in layers — CODE:
LAYER 0
col 0 col 1 col 2 col 3
row 0 12 15 18 20
row 1 21 24 27 30
row 2 32 35 38 40

LAYER 1
col 0 col 1 col 2 col 3

row 0 42 45 48 50
row 1 51 54 57 60
row 2 62 65 68 70 — Worked example: find scores[1,2,1] .

6. Three dimensions need three loops — CODE:

```
for (int layer = 0;
layer < scores.GetLength(0);
layer++)
{
for (int row = 0;
row < scores.GetLength(1);
row++)
{
for (int column = 0;
column < scores.GetLength(2);
column++)
{
Console.WriteLine(
scores[layer,row,column]);
}
}
}
```

We Do — guided CHECK

1. Read the array

2. Locate the element

3. Say it before you write it
Locate 57

4. Build the declaration
OCR's mark scheme awards the declaration for identifying an array/string named towns and setting the two dimensions to 8 and 3.

You Do — source mark schemes

1. [No safely extracted source model: staff must check]

2. [No safely extracted source model: staff must check]

3. [No safely extracted source model: staff must check]

4. [No safely extracted source model: staff must check]

5. [No safely extracted source model: staff must check]

6. [No safely extracted source model: staff must check]

7. [No safely extracted source model: staff must check]

8. [No safely extracted source model: staff must check]

9. [No safely extracted source model: staff must check]

10. [No safely extracted source model: staff must check]

Exit Ticket — source answer

1. True. The three commas in int[,,] specify a three-dimensional C# array.